BRAC UNIVERSITY

Department of Computer Science and Engineering

CSE 341: Microprocessors Semester: Summer 2026

Assignment 03 Total Marks: 40

Question 1 [18 marks]

A space station named **"The Starry Station"** is hosting a special intergalactic broadcast, **"When the Star Gossips"**, featuring **Lee Min-ho.** The station's central system is controlled by an **Intel 8086 microprocessor** and uses a fictional enhanced Programmable Interrupt Controller called **PIC-X.** To keep the station under control, PIC-X must efficiently handle the incoming interrupt requests.

Unlike the conventional 8259A, PIC-X can handle **20 different interrupt requests** at a certain time and can identify **65,536** different **interrupt types.** PIC-X is currently operating in **fixed-priority mode.**

a) Determine the number of bits required for the IRR, ISR, and IMR of PIC-X. [1]

b) At T = 32ns, IR6 receives an interrupt. The CPU will take 50ns to serve that interrupt. However, at T = 45ns, an interrupt occurs at IR5 which requires 20ns to get served. At T = 52ns, two interrupt requests occur at IR7 and IR0 respectively. Each of the interrupts will require 37ns to get served. Deduce the value of T when the interrupt at IR6 is served completely. [3]

c) Explain in detail how the values of the IRR, ISR, and IMR registers will change according to the above requests. [3]

d) Explain in detail the actions taken by 8086 when responding to an interrupt request in the INTR pin. [3]

e) Why does 8086 clears IF and TF while serving an interrupt? [2]

f) Differentiate between Hardware and Software interrupts in details.[3]

g) Deduce with reasoning how many data bus lines of the PIC "X" would be connected to the 8086 data bus? [1]

Question 2 [10 marks]

An 8086 system uses cascaded 8259A PICs for hardware interrupts. The system must handle **25 interrupt sources.** Due to faults, **IR3 and IR7** are unusable on all PICs. Slaves connect to the master sequentially from Master IR0, skipping faulty pins.

The values of the registers for executing the below code snippet:

SS = A000h, SP = **FFFA**h, CS = 3000h, DS = 2000h, ES = 1000h

The following instructions are executed sequentially:

Instruction 1:	MOV AX, EF17H
Instruction 2:	MOV BL, 13H
Instruction 3:	DIV BL
Instruction 4:	MOV VAR, AL

Assume that the interrupt service routine for the interrupt generated during execution of the above instructions starts at physical address **8B62AH**, with a fixed CS of **8A00H**.

a) Analyze with **proper justification** the given instructions and determine **which interrupt is generated** during their execution and its **interrupt type. [1.5 marks]**

b) Using the information provided about the ISR, calculate the **IP** of the ISR. Then determine the **four IVT memory addresses** and corresponding **hexadecimal data values** required to install this interrupt. **[2.5 marks]**

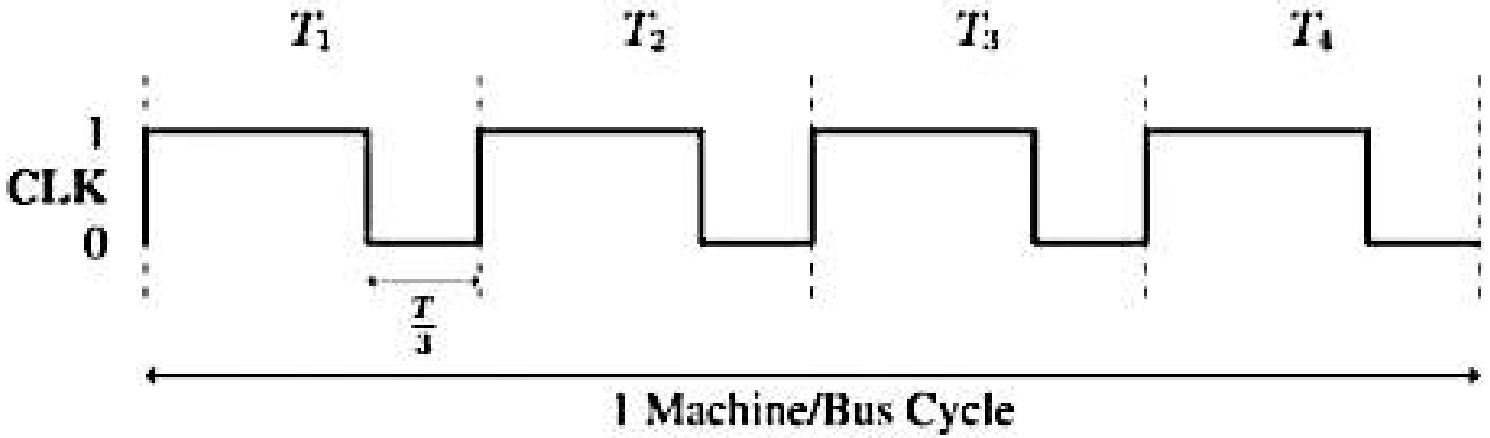
c) The instruction that generates the interrupt **starts at** physical address **31122H** and is **2 bytes long.** Deduce the hexadecimal values pushed onto the stack at physical addresses **AFFF4H through AFFF7H** when the interrupt occurs. Assume the **FLAGS** are pushed to **AFFF8H–AFFF9H. [3 marks]**

d) Determine the **minimum** number of PICs required to handle the **25 sources** including **master and slave(s). [1 marks]**

e) Each slave simultaneously receives requests on **IR0** and **IR1.** Master **IMR = 00010001, ISR = 10000000.** Slave **IMR = 10010010, ISR = 00000000 (The bits are ordered from right to left).** Identify which slave's interrupt is serviced first and explain the priority/mask logic. **[2 marks]**

Question 2 [12 marks]

Consider the following 8086 assembly instruction: **MOV BX, [3001H].** The 8086 processor operates with T_{on} **= 55.5ns.** The memory connected to the 8086 is slow memory; therefore, one wait state (T_w) is inserted during each bus cycle while executing the instruction. The processor clock is generated from the waveform shown below.



a) Calculate the **frequence(MHz)** at which the processor is operating. [2]

b) Calculate the length of **one instruction cycle** to execute the given instruction. [3]

c) Determine the **Duty Cycle** for the given case. [2]

d) Explain with **proper reasoning** the values of pins **A0** and **BHE'** during the execution of the given instruction. [2]

e) Illustrate the **timing diagram** for the given instruction. Include the following pins in your diagram: **CLK, AD0-AD15, A16-A19/S3-S6, ALE, RD'/WR', M/IO', READY, DEN',** and **DT/R'.** [3]